ASSIGNMENT 1

Resume Screening Using Natural Language Processing

Detailed Assignment Report

NLP Application: Resume Screening / Applicant Tracking System (ATS)

Evaluation of real-world recruitment tools and adaptation for Marathi

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1. Introduction

Recruitment is one area where Natural Language Processing (NLP) can be used to reduce the amount of manual work involved in screening candidates. A recruiter may receive many resumes for a single opening, and reviewing every resume manually can take considerable time. Resume-screening systems attempt to help with this first stage by extracting information from resumes, identifying skills and qualifications, and comparing candidates with job requirements.

This assignment studies resume screening as a real-world NLP application. The practical work involved creating a job opening, using a candidate resume, testing resume parsing, checking ATS/job-description matching, and using an AI-based candidate evaluation tool. The outputs from the different tools were then compared to understand what each system does well and where automated screening can make mistakes.

The assignment also considers how the same application could be adapted for Marathi. This is important because a multilingual recruitment system cannot depend only on exact English keywords. Marathi resumes may contain Marathi text, English technical terms, mixed Marathi-English sentences, different scripts, and regional language variations.

2. NLP Application: Resume Screening

The selected NLP application is Resume Screening / Applicant Tracking System (ATS). The main objective is to automatically process resume text, extract useful candidate information, compare that information with a job description, and assist recruiters in shortlisting candidates.

2.1 Main NLP Tasks Involved

- Text extraction from an uploaded resume.
- Resume parsing and conversion of unstructured information into structured fields.
- Identification of skills, qualifications, job titles, organizations, and work experience.
- Keyword and semantic matching between a resume and a job description.
- Candidate scoring and ranking.
- Identification of missing skills or experience.
- Generation of explanations that help a recruiter understand the screening result.

The practical experiment shows that these tasks are distributed across different tools. One tool may be stronger at parsing, while another focuses on ATS matching and another provides a broader candidate-fit assessment.

3. Tools and Practical Evaluation

3.1 Zoho Recruit

Zoho Recruit was used to create and manage the recruitment workflow and to test resume parsing. The candidate resume was uploaded and the system extracted structured information from the document.

Observed result: Resume parsing accuracy: 100%.

The parsing result showed that information such as occupation/title, company, summary, and work duration could be extracted and presented in a structured form. This makes the resume easier for a recruiter to review.

3.2 Jobscan

Jobscan was tested using a Junior Data Analyst job description. It compared the candidate resume with the job requirements and evaluated ATS compatibility, searchability, hard skills, and other matching factors.

Observed result: Match Rate: 67%.

The result indicates that the resume had a reasonable level of alignment with the job but still had areas that could be improved for ATS matching. The result is a screening signal rather than a final statement about the candidate's ability.

3.3 HyreTech

HyreTech was used for AI-based candidate evaluation. The tool provided an overall score, a recommendation, technical qualification assessment, work-experience assessment, and information about matching and missing requirements.

Observed result: Overall Score = 71/100; Verdict = Consider; Technical Qualifications = 78; Work Experience = 64.

The evaluation was useful because it did not rely only on a single number. It also highlighted strengths and gaps, making it easier to see why the candidate was considered but not treated as an automatic match.

4. Input Data Used in the Experiments

4.1 Job Opening

The main recruitment scenario used a Junior Data Analyst job opening. The role was configured as a full-time position with an expected experience level of 0–1 year in an IT Services environment.

- Python
- SQL
- Excel
- Power BI
- Data Analysis
- Statistics
- Data Visualization
- Pandas
- NumPy

The responsibilities included analyzing datasets, preparing reports, creating dashboards, cleaning and organizing data, identifying trends and patterns, and communicating analytical insights.

4.2 Candidate Resume

The candidate used in the screening exercise was Jake Ryan. The resume contains education, work experience, projects, and a technical-skills section. The technical background includes Python,

SQL/PostgreSQL, JavaScript, React, Flask, FastAPI, Docker, Git, TravisCI, pandas, NumPy, Matplotlib, and other technologies.

The resume also includes projects involving REST APIs, PostgreSQL, GitHub data, Flask, React, Docker, Celery, and Redis. These details are important when interpreting the automated scores because they show that the candidate has substantial technical exposure, while the resume is not positioned purely as a data-analyst resume.

5. Task (i): Testing a Real-World Tool

5.1 (a) Data Given to the Tool

The main input consisted of a job description and a candidate resume. The job description provided the target role, skills, experience expectations, responsibilities, and educational requirements. The resume provided the candidate's education, work history, projects, and technical skills.

Input	Details
Target role	Junior Data Analyst
Employment	Full-time
Experience	0–1 year
Key skills	Python, SQL, Excel, Power BI, Data Analysis, Statistics, Data Visualization, Pandas, NumPy
Candidate	Jake Ryan
Resume information	Education, work experience, projects, technical skills and related details

5.2 (b) How the System Responded

Zoho Recruit parsed the resume and displayed extracted candidate information. The parsing stage reported 100% parsing accuracy.

Jobscan compared the resume against the Junior Data Analyst requirements and reported a 67% match rate. It also highlighted ATS/searchability factors and areas where the resume could be better aligned with the job.

HyreTech provided an overall score of 71/100 and a 'Consider' verdict. It reported a technical qualification score of 78 and a work-experience score of 64. It also identified matching skills and missing or weaker areas.

5.3 (c) Where the System Succeeded

- The resume was successfully parsed and converted into structured information.
- The tools identified several genuine technical matches from the candidate's resume.
- The systems were able to identify gaps rather than simply returning a yes/no result.
- The results made it possible to compare technical capability with role-specific experience.
- The tools reduced the amount of manual effort required for the initial screening stage.

5.4 Where the System Can Fail or Be Limited

- A keyword match does not prove that the candidate has deep practical experience.
- Different wording for the same skill can reduce an ATS score.
- A resume may contain a technology without explaining the context in which it was used.
- Resume formatting and section headings can influence machine readability.
- Automated scores cannot completely measure real-world ability, communication, or problem-solving.

5.5 Non-English / Indian-Language Input

The supplied practical screenshots primarily demonstrate English job and resume inputs. Therefore, the practical experiment does not provide direct measured evidence of the tools' performance on a Marathi resume. This is a limitation of the experiment. The Marathi section of this report therefore proposes the resources and testing steps required for a genuine Marathi adaptation rather than claiming that Marathi performance was already measured.

6. Task (ii): Comparing Multiple Tools

The same recruitment scenario was examined through different tools. Their outputs were different because each tool focuses on a different part of the screening process.

Tool	Main purpose	Observed output	Interpretation
Zoho Recruit	Resume parsing	100% parsing accuracy	The resume information was extracted successfully.
Jobscan	ATS/searchability and job-description matching	67% match	The resume has relevant content but can be better aligned with the target job.
HyreTech	Candidate scoring and reasoning	71/100 – Consider	Technical fit is relatively strong, while role-specific experience has gaps.

6.1 (a) Differences in Outputs

- Zoho Recruit mainly answers whether resume information can be parsed and structured.
- Jobscan focuses on how closely the resume matches the job description and how well it may perform in ATS screening.
- HyreTech gives a broader candidate-fit assessment and provides explanations for strengths and gaps.
- Because the objectives are different, the scores are not expected to be identical.

6.2 (b) Types of Errors

- Keyword mismatch: a relevant skill may not be recognized if the resume uses different wording.
- Context loss: the system may see a technology but not understand the depth of the candidate's experience with it.

- Experience interpretation: total professional experience is not necessarily the same as experience in the target role.
- Parsing limitations: unusual formatting or document structures can affect extraction.
- Semantic limitations: matching words does not guarantee matching ability.
- Recommendation limitations: an automated score or verdict should not be treated as the final hiring decision.

6.3 (c) Cultural or Regional Misunderstanding

The screenshots do not contain a controlled Marathi or other Indian-language experiment, so a measured cultural error rate cannot be claimed. However, a multilingual screening system can face difficulties with local-language expressions, names, regional terminology, mixed-language writing, and different ways of describing education or work.

For Marathi recruitment, the system should therefore be evaluated using Marathi-only and Marathi-English mixed resumes from different regions and writing styles. Human recruiter evaluation should be used as a reference when checking whether the automated output is correct.

7. Detailed Error Analysis

The experiments showed that automated resume-screening systems can process resumes quickly, but their results may be affected by keyword matching, resume structure, missing context, and interpretation of experience. The following sections explain the major error types in more detail.

7.1 False Positives

A false positive occurs when the system considers a candidate suitable because relevant keywords or information are present, even though the candidate may not fully satisfy the job requirement.

In the HyreTech experiment, several backend technologies such as Python, Flask, FastAPI, PostgreSQL, Docker, Celery, and Redis were identified as matching skills. However, the tool also identified that the resume did not clearly demonstrate three or more years of specific backend-service development.

This shows that the presence of relevant technical skills does not necessarily mean that the candidate has sufficient practical experience for the position. A candidate may have used a technology in an academic project or personal project without having equivalent professional experience.

7.2 False Negatives

A false negative can occur when a candidate possesses relevant knowledge or experience, but the system fails to recognize it because the information is written differently in the resume.

ATS-oriented systems may depend heavily on keywords. A candidate may describe a skill using a related term, abbreviation, synonym, or different wording. The system may then assign a lower match score even when the underlying skill is relevant.

This is especially important for multilingual resumes. A candidate may describe a skill in Marathi while the job description uses the corresponding English term.

7.3 Keyword Matching Limitations

Jobscan reported a 67% match rate for the Junior Data Analyst resume. The tool also identified ATS-related issues involving searchability and section headings.

This demonstrates that resume screening can be influenced by how information is written and organized rather than only by the candidate's actual capabilities. Using the exact job title, standard section headings, and relevant skill names can improve machine readability, but keyword optimization should not replace honest representation of experience.

7.4 Context and Experience Limitations

A resume may mention a technology without providing enough information about how extensively it was used.

"Python, Docker and PostgreSQL"

This list does not indicate whether the candidate used these technologies in a major production system, an academic project, an internship, or a small personal project.

HyreTech partially addressed this issue by distinguishing between listed skills and evidence of professional backend experience. It identified the lack of clearly documented backend-specific experience as a gap.

7.5 Resume Parsing Limitations

Zoho Recruit reported 100% parsing accuracy for the tested resume. Successful extraction, however, does not necessarily mean that the system has understood the complete meaning of the resume.

Resume parsing mainly converts unstructured information into structured fields. A separate matching or semantic-analysis process is required to determine whether the extracted information actually satisfies a specific job requirement.

Therefore, a 100% parsing result should not be interpreted as a 100% candidate-suitability result.

7.6 Overall Error Analysis

The experiments demonstrate that automated resume screening should be treated as a decision-support system rather than a complete replacement for human recruiters.

- Zoho Recruit focuses on structured resume parsing.
- Jobscan emphasizes ATS compatibility and job-description matching.
- HyreTech provides candidate scoring and explanations of matching and missing qualifications.
- Combining structured parsing, semantic skill matching, experience analysis, and human review can produce a more reliable recruitment process.

8. Adaptation of Resume Screening for Marathi

8.1 Indian Language Selected

The selected Indian language is Marathi. The objective is to adapt an automated resume-screening system so that it can understand Marathi resumes and Marathi-English mixed resumes and match them with relevant job descriptions.

A Marathi-enabled system could make automated recruitment more accessible to candidates who prefer writing their resumes in Marathi while also improving recognition of skills and qualifications expressed using Marathi or a mixture of Marathi and English.

8.2 Required Linguistic Resources

8.2.1 Marathi Resume Corpus

A Marathi resume corpus would be required for training and testing. It should contain education, skills, work experience, job titles, projects, and certifications.

"मला Python, SQL आणि Data Analysis मध्ये अनुभव आहे."

The system should identify Python, SQL, and Data Analysis as relevant skills.

8.2.2 Marathi Job Description Corpus

A collection of Marathi job descriptions would help the system learn how job requirements are expressed in Marathi.

"उमेदवाराला Python आणि SQL चे ज्ञान असणे आवश्यक आहे."

The system should associate this requirement with the corresponding skills in a candidate's resume.

8.2.3 Marathi POS Tagger

A Part-of-Speech (POS) tagger can help identify the grammatical role of words in Marathi sentences. This can help the system process resume sentences and distinguish important information from general text.

8.2.4 Named Entity Recognition (NER)

A Marathi NER system can be used to identify:

- Candidate names
- Organizations
- Colleges and universities
- Locations
- Job titles
- Educational qualifications
- Skills

This can help convert an unstructured Marathi resume into structured candidate information.

8.2.5 Marathi Language Model

A Marathi language model can help the system understand the context and meaning of resume statements rather than relying only on exact keyword matching.

"डेटा विश्लेषणाचा अनुभव आहे."

This should be recognized as related to "Experience in Data Analysis."

8.2.6 Marathi-English Lexicon and Parallel Data

Many technical resumes use English words even when the rest of the resume is written in Marathi. A Marathi-English vocabulary and parallel dataset can help map Marathi expressions to their English equivalents and connect the same skill across languages.

8.3 Socio-Linguistic Challenges

8.3.1 Code-Switching

Candidates may use Marathi and English in the same sentence.

"मी Python आणि SQL वापरून data analysis projects केले आहेत."

The system should still correctly identify Python, SQL, and Data Analysis as relevant skills.

8.3.2 Dialect Variations

Marathi is used across different regions and may contain variations in vocabulary and expressions. A resume-screening system should avoid treating regional differences in wording as missing skills or qualifications.

The system should therefore be trained and evaluated using text from different Marathi-speaking regions.

8.3.3 Script Variations

Marathi is normally written using the Devanagari script, but users may also type Marathi using Roman/Latin script.

Devanagari: "मला Python मध्ये अनुभव आहे."

Roman Marathi: "Mala Python madhye anubhav aahe."

The adapted system should ideally recognize that both forms express the same meaning.

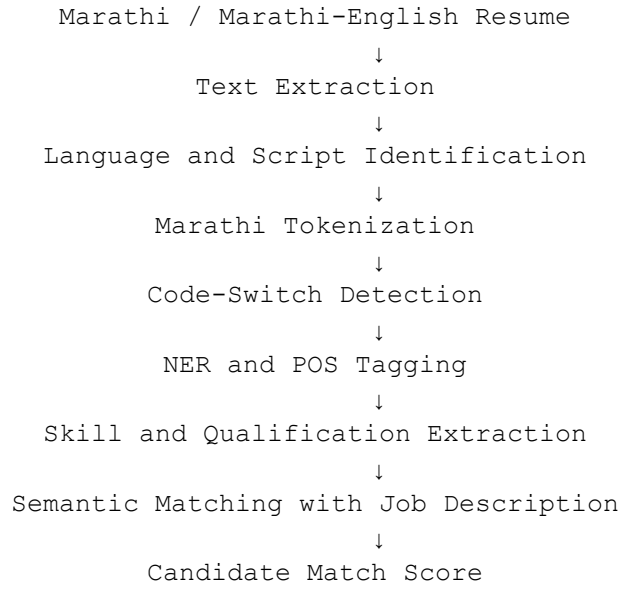
8.3.4 English Technical Terms

Technical resumes frequently contain English terminology even when the surrounding text is Marathi.

"मला Machine Learning, Python आणि Data Visualization मध्ये experience आहे."

The system should identify Machine Learning, Python, and Data Visualization as technical skills.

8.4 Proposed Marathi Resume-Screening Pipeline



↓
Candidate Ranking

8.5 Concrete Steps for Adaptation

1. Collect Marathi resumes and Marathi job descriptions.
2. Annotate skills, education, organizations, job titles, projects, and experience.
3. Add Marathi language-processing resources such as POS tagging, NER, and language models.
4. Build a Marathi-English skill vocabulary.
5. Add code-switching detection for Marathi-English resumes.
6. Support both Devanagari and Roman Marathi text.
7. Train the matching system using Marathi and Marathi-English examples.
8. Test the system using resumes from different regions and writing styles.
9. Compare Marathi system results with the original English resume-screening system.
10. Use human recruiter evaluation to identify incorrect matches and improve the model.

8.6 Expected Benefits

A Marathi-enabled resume-screening system would make automated recruitment more accessible to candidates who prefer writing resumes in Marathi. It could reduce dependence on exact English keywords and improve recognition of skills and qualifications expressed using Marathi or Marathi-English mixed language.

However, human review would still be important because automated systems may misunderstand regional language variations, mixed-language expressions, dialect differences, or context-specific descriptions.

9. Three-Point Candidate Analysis

9.1 Technical Fit

The candidate shows a good technical foundation. The resume includes Python, SQL/PostgreSQL, Flask, FastAPI, Docker, Git, pandas, NumPy, Matplotlib, and other software-development tools. The HyreTech technical qualification score of 78 supports the view that the technical background is one of the candidate's stronger areas.

9.2 Experience Fit

The main weakness is direct alignment with the target role. The candidate's experience is more strongly presented around research, IT support, AI research, and software projects than direct professional data-analyst work. This helps explain why experience-related evaluation is weaker than the technical evaluation.

9.3 Overall Hiring View

The combined evidence suggests that the candidate should be considered rather than automatically rejected. The 67% Jobscan match and 71/100 HyreTech score point to a candidate with useful technical skills but with resume-positioning and role-specific experience gaps. A practical data-analysis test or interview would help verify whether the candidate can perform the required work.

10. Overall Findings

Area	Finding
Resume parsing	Successful; Zoho Recruit reported 100% parsing accuracy.
ATS/job matching	Moderate alignment; Jobscan reported 67%.
AI candidate evaluation	Candidate received 71/100 and a 'Consider' verdict.
Technical capability	Relatively strong, with several relevant programming/data technologies.
Role-specific experience	Less clearly aligned with the target data-analyst role.
Main screening risk	Keyword and context limitations can distort automated scores.
Marathi adaptation	Requires language resources, multilingual handling, and testing across regional variations.
Final approach	Use automated tools as decision support together with human review.

11. Limitations of the Practical Experiment

The practical work demonstrates the use of real recruitment tools, but it is not a controlled scientific benchmark. The tools have different purposes, the screenshots represent specific test cases, and the practical experiment did not include a controlled Marathi resume/job-description evaluation.

- The observed scores apply to the tested resume and job configurations.
- Different tools use different scoring methods, so their numerical values are not directly interchangeable.
- The screenshots do not establish a measured Marathi accuracy or cultural-bias rate.
- Human recruiter evaluation was not replaced by an automated score.
- The Marathi section is an adaptation proposal based on the assignment requirements, not a completed Marathi model implementation.

12. Conclusion

This assignment demonstrates how NLP can support several stages of recruitment, including resume parsing, ATS matching, skill extraction, candidate scoring, and identification of experience gaps.

The three tools provided different but complementary views of the candidate. Zoho Recruit demonstrated effective resume parsing with a reported 100% parsing accuracy. Jobscan provided an ATS-oriented match score of 67%, while HyreTech provided a broader evaluation of 71/100 with a 'Consider' recommendation.

The error analysis shows why these scores should not be treated as final hiring decisions. Keyword matching can miss relevant experience, a listed skill does not prove practical expertise, and resume parsing does not equal semantic understanding. Human review is therefore important.

For Marathi, an effective adaptation would require Marathi resume and job-description corpora, POS and NER resources, Marathi language models, Marathi-English lexical resources, code-switching handling, support for Devanagari and Roman Marathi, and evaluation across regional language variations. Combining these resources with semantic matching and human review could make resume screening more inclusive and reliable.

13. Evidence and Screenshot Placement

For the final submission, the screenshots collected during the practical work should be inserted in the report near the corresponding sections. The recommended order is:

11. Job opening and required skills.
12. Resume upload / resume parsing result showing the parsing output.
13. Candidate profile and extracted information.
14. Jobscan job description and 67% match result.
15. HyreTech 71/100 score and 'Consider' result.
16. HyreTech technical qualifications, matching skills, missing requirements, and gap analysis.

These screenshots provide practical evidence for the results discussed in the report. The final report should retain the original screenshots rather than relying only on written descriptions.